

CARLOS DANIEL TORRES MACÍAS

Intelligent Computing Engineering · Data Analysis · Software Development

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PROFESSIONAL PROFILE

Tenth-semester Intelligent Computing Engineering student (graduating June 2026) with a strong background in data analysis, machine learning, software development, and systems administration. Experienced in building data pipelines, training and evaluating classification models, and developing full-stack applications. Skilled at translating complex problems into concrete technical solutions in both research and applied project environments. Additional experience teaching programming at the high school level.

EDUCATION

Universidad Autónoma de Aguascalientes, Aguascalientes, Mexico 2021 – Jun 2026

B.S. in Intelligent Computing Engineering (in progress)

- Relevant coursework: Statistics · Databases · Data Analysis and Machine Learning · Deep Learning · Reinforcement Learning · Metaheuristics · Data Mining · Pattern Recognition · Networking · Algorithms and Data Structures · Operating Systems.

SQL TOTAL: Master Databases from 0 to Advanced, Udemy 2026

TECHNICAL SKILLS

Languages: Python · SQL · PL/SQL · Java · C / C++ · JavaScript · HTML/CSS

Data & ML: pandas · numpy · scikit-learn · XGBoost · SHAP · data mining

Visualization / BI: Power BI · Google Data Studio · matplotlib · seaborn

Statistics: Descriptive and inferential statistics · regression · distribution analysis · classification metrics

Databases: MySQL · PostgreSQL · MariaDB · relational modeling · advanced queries

Development: Git / GitHub · React · REST APIs · FastAPI · Docker · Linux · Scrum / Kanban

Tools: VSCode · Jupyter · Advanced Excel · BIND (DNS) · Proxmox

Languages: Spanish (native) · English B2

Soft skills: Analytical thinking · Clear technical communication to non-specialist audiences · Autonomy and continuous self-directed learning

PROJECTS

Hybrid Phishing Detection System — Undergraduate Thesis (in progress) 2025 – 2026

- Late-fusion architecture with metadata gating: Random Forest over 32 technical metadata features combined with TF-IDF + Logistic Regression on email body text. Gating mechanism short-circuits to phishing classification when metadata confidence exceeds threshold, reducing unnecessary NLP inference.
- End-to-end pipeline over 164,563 MD5-deduplicated records from 7 public datasets (stratified 70/15/15 split). Text submodel ROC-AUC: 0.9988; metadata submodel: 0.9121.
- SHAP TreeExplainer for interpretability: explanations matched human expert judgment in 94% of evaluated cases; full explainability latency under 30 ms.
- REST microservice (FastAPI + Docker) with schema-versioned model artifacts, Pydantic v2 validation, and CI via GitHub Actions.
- Stack: Python · scikit-learn · SHAP · FastAPI · Docker · pandas · Git LFS

Distributed Moodle Cluster on Raspberry Pi — UAA Miniproject 2025

Role: responsible for the federated database layer (MariaDB)

- Design and implementation of federated replication in MariaDB across five Raspberry Pi 4B nodes, including replication user configuration and my.cnf parameters.
- Troubleshooting synchronization and connection saturation issues under load tests (50, 100, and 200 concurrent users with JMeter and Locust).
- Service stability validation: response times below 500 ms with 100 simultaneous users in a low-cost, low-power architecture (~5 W per node).
- Stack: MariaDB · NFS · Apache (mod_proxy_balancer) · Linux · Raspberry Pi.

Home Server — Personal Lab 2023 – 2024

- CentOS server with user administration, task automation, SSH/VPN remote access, DNS server with BIND, and system monitoring.
- Hands-on experience with Proxmox for development environment virtualization.

EXPERIENCE

Preparatoria Marsella, Aguascalientes Oct 2025 – Present

Programming Instructor

- Taught Python, Networking, and Digital Culture to high school groups.
- Designed activities focused on analytical thinking, computational logic, and problem-solving.

Apoyo Financiero – Conexia, Aguascalientes Mar 2025 – Present

Bilingual Agent

- Performance metrics tracking and KPI reporting via CRM.

Home Bed, Aguascalientes Jul 2023 – Mar 2025

Telephone Agent

- Data analysis for customer prioritization and operational reporting with advanced Excel.

Financiera Independencia – Conexia, Aguascalientes Feb 2022 – Jan 2023

Telephone Agent

- CRM operations logging and credit portfolio indicator tracking.